

ADS-509 – Team Project Status Update Form

Fill out this form and submit it by the end of Module 4 in Canvas.

Team Number: **Team 1**

Team Leader/Representative: **Zadafiya Bhrami**

Full names of team members:

1. Zadafiya Bhrami
2. Amarnath Sreekara

Title of your Applied Text Mining project: **Financial News Sentiment Classification using Fine-tuned FinBERT**

Short description of your project and objectives:

This project develops a text classification pipeline to automatically detect sentiment (Positive, Negative, Neutral) in financial news headlines. General-purpose sentiment tools like VADER perform poorly on financial text because domain-specific language (e.g., "impairment charges", "missed guidance") lacks clear polarity in standard lexicons.

Our objectives are: (1) perform exploratory data analysis on Financial PhraseBank to understand class distribution, vocabulary patterns, and text structure; (2) establish a VADER rule-based baseline; (3) fine-tune FinBERT — a BERT model pre-trained on financial corpora — using the HuggingFace Transformers library; and (4) evaluate and compare model performance using accuracy, macro F1, and per-class F1 scores.

Name of your selected dataset: **Financial PhraseBank**

Description of your selected dataset (data source, number of variables, size of dataset, etc.):

Source: Publicly available on Kaggle (kaggle.com/datasets/ankurzing/sentiment-analysis-for-financial-news).

Originally published by Malo et al. (2014) as part of academic research on financial sentiment.

Size: 4,840 English-language sentences extracted from financial news articles.

Variables: 2 columns — (1) sentiment label: Positive / Negative / Neutral;
(2) text: the raw headline or sentence.

Class dist.: Neutral ~59.5% (2,879 samples), Positive ~28.2% (1,363), Negative ~12.5% (604).
Dataset is imbalanced; Negative class is underrepresented.

Format: CSV file (all-data.csv), UTF-8 / Latin-1 encoded, no missing values after cleaning.

How many times have your members met in the last two weeks? 3-4 times

List the specific contributions that each team member is providing for the Final Team Project in the table below.

- **NOTE:** ALL students on the team should contribute equally to the Final Team Project.

Team Member 1 Zadafiya Bhrami	Team Member 2 Amarnath Sreekara
• Dataset sourcing and download (Financial PhraseBank from Kaggle) • Data cleaning and preprocessing (whitespace normalisation, label encoding, train/val/test split) • Exploratory Data Analysis: class distribution chart, sentence length histogram • Word cloud visualisations • VADER baseline implementation and evaluation	• FinBERT model setup using HuggingFace Transformers (tokenisation, dataset conversion) • Model fine-tuning: TrainingArguments configuration, Trainer loop, early stopping • Model evaluation: confusion matrix, classification report, per-class F1 analysis • Figure generation: model comparison bar chart, per-class F1 chart • Error analysis: misclassified example review and pattern identification